

Capstone Project

# Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

|              |                                                     |
|--------------|-----------------------------------------------------|
| Company      | Bluestock Fintech Pvt. Ltd.                         |
| Domain       | Mutual Fund / Fintech                               |
| Project Type | Individual Capstone (1 Week)                        |
| Duration     | 7 Working Days   ~50-55 Hours                       |
| Data Source  | AMFI India (Public), mfapi.in, NSE/BSE Public Data  |
| Technologies | Python, SQL, Power BI / Tableau, Pandas, Matplotlib |
| Submission   | PDF Report + Dashboard + GitHub Repository          |
| Prepared By  | Intern / Data Analyst — Bluestock Fintech           |
| Date         | June 2026                                           |

40 Schemes

Real Fund Schemes

10 Datasets

Provided

87K+ Rows

Transaction Data

4.5 Yrs

NAV History

All data in this project is sourced from publicly available information published by AMFI India , NSE, BSE and open APIs (mfapi.in). This project is for educational purposes only and does not constitute financial advice. Mutual Fund Investments are subject to market risks.

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## 1. Project Overview

This project builds an end-to-end mutual fund analytics platform for Bluestock Fintech. Starting from raw AMFI/SEBI data, the pipeline cleans and structures 10 datasets covering 40 mutual fund schemes, loads them into a SQLite star-schema database, performs exploratory and performance analysis in Jupyter notebooks, and delivers advanced analytics including Value-at-Risk modelling, investor cohort retention, and a risk-based fund recommender system.

## Technology Stack

| Component       | Technology                  | Version           |
|-----------------|-----------------------------|-------------------|
| Language        | Python                      | 3.11+             |
| Data processing | pandas, numpy               | 3.0 / 2.4         |
| Visualisation   | matplotlib, seaborn, plotly | 3.10 / 0.13 / 6.7 |
| Database        | SQLite3 (built-in)          | 3.x               |
| Statistics      | scipy                       | 1.17              |
| Notebooks       | JupyterLab                  | 4.5               |
| Live NAV        | mfapi.in REST API           | —                 |
| Package mgmt    | pip + requirements.txt      | —                 |

## 2. Repository Structure

The repository follows the recommended capstone folder layout. All scripts use `pathlib.Path` for cross-platform paths; no absolute paths are hard-coded in production code.

```
mutual-fund-dashboard/  
├── data/  
│   ├── raw/ ← 10 source CSVs + 6 live NAV CSVs from mfapi.in  
│   ├── processed/ ← Cleaned CSVs + alpha_beta.csv, fund_scorecard.csv,  
│   │   tracking_error.csv, var_analysis.csv  
│   └── notebooks/  
│       ├── EDA_Analysis.ipynb  
│       ├── Performance_Analysis.ipynb  
│       └── 05_advanced_analytics.ipynb  
├── scripts/  
│   ├── etl_pipeline.py ← Single-entry ETL: ingest → clean → load  
│   └── sql/  
│       ├── schema.sql ← Star schema DDL  
│       ├── queries.sql ← 10 analytical queries  
│       └── load_db.py ← FK-aware loader  
├── reports/  
│   ├── charts/ ← 12 saved charts (PNG + HTML)  
│   ├── data_dictionary.md  
│   ├── data_quality_summary.md  
│   ├── data_ingestion.py ← Anomaly scanner  
│   ├── data_cleaning.py ← Per-table cleaning rules  
│   ├── live_nav_fetch.py ← mfapi.in fetcher  
│   ├── validate_amfi_codes.py  
│   ├── requirements.txt  
└── .gitignore ← Includes *.db
```

### 3. Dataset Description

10 source files covering 40 mutual fund schemes, 2022–2026

| File                         | Rows (approx) | Key Columns                                                  | Notes                                                              |
|------------------------------|---------------|--------------------------------------------------------------|--------------------------------------------------------------------|
| 01_fund_master.csv           | 40            | amfi_code, scheme_name, fund_house, category, risk_grade     | Dimension table; primary key                                       |
| 02_nav_history.csv           | 46 000        | amfi_code, date, nav                                         | Forward-filled for holidays via <code>resample('D').ffill()</code> |
| 03_aum_by_fund_house.csv     | 500+          | date_id, fund_house, aum_crore, aum_lakh_crore               | Units explicit in column names                                     |
| 04_monthly_sip_inflows.csv   | 40+           | date_id, sip_inflow_crore, active_sip_accounts_crore         | Industry-level monthly aggregates                                  |
| 05_category_inflows.csv      | 200+          | date_id, category, net_inflow_crore                          | Equity/Debt/Hybrid split                                           |
| 06_industry_folio_count.csv  | 40+           | date_id, total_folios_crore, equity_folios_crore             | Total folios approaching 20 Cr                                     |
| 07_scheme_performance.csv    | 40            | amfi_code, return_1yr_pct, return_3yr_pct, expense_ratio     | Expense ratio validated: 0.1–2.5% range                            |
| 08_investor_transactions.csv | 5 000         | investor_id, transaction_type, amount_inr, state, kyc_status | SIP / Lumpsum / Redemption normalised                              |
| 09_portfolio_holdings.csv    | 400+          | amfi_code, stock_name, sector, weight_pct                    | Top holdings per scheme                                            |
| 10_benchmark_indices.csv     | 3 000+        | date, index_name, close_value                                | NIFTY50/100/500, BSE SmallCap, CRISIL Liquid/Gilt                  |

#### Live NAV Data (mfapi.in)

Six large-cap schemes were also fetched live from the MFAPI REST endpoint (<https://api.mfapi.in/mf/{code}>). The fetcher (`live_nav_fetch.py`) stores full historical NAV as individual CSVs in `data/raw/`.

| Scheme              | AMFI Code |
|---------------------|-----------|
| HDFC Top 100 Direct | 125497    |
| SBI Bluechip        | 119551    |
| ICICI Pru Bluechip  | 120503    |
| Nippon Large Cap    | 118632    |
| Axis Bluechip       | 119092    |
| Kotak Bluechip      | 120841    |

## 4. D1 — ETL Pipeline

*scripts/etl\_pipeline.py | data\_ingestion.py | data\_cleaning.py*

The ETL is implemented in two complementary layers. **data\_ingestion.py** is an anomaly scanner — it profiles every CSV for missing values, duplicates and empty columns, and reports findings. **data\_cleaning.py** applies per-table transformation rules. **scripts/etl\_pipeline.py** is the unified entry-point that chains ingest → clean → SQLite load in a single python `scripts/etl_pipeline.py` call.

### Pipeline Functions

| Function                                  | File                           | Description                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ingest_raw(raw_dir)</code>          | <code>etl_pipeline.py</code>   | Globs all *.csv, normalises column names (strip/lower/replace spaces), returns {stem: DataFrame} dict. Logs each file loaded.                                                                                              |
| <code>basic_clean(tables, out_dir)</code> | <code>etl_pipeline.py</code>   | Trims string columns, drops fully-empty columns, writes *_clean.csv to data/processed/.                                                                                                                                    |
| <code>load_sqlite(processed, db)</code>   | <code>etl_pipeline.py</code>   | Iterates *_clean.csv, writes each as a SQLite table with if_exists='replace'. Creates db parent dir.                                                                                                                       |
| <code>main()</code>                       | <code>etl_pipeline.py</code>   | Detects repo root via Path.cwd() + name check. Calls the three functions above in order.                                                                                                                                   |
| (per-table logic)                         | <code>data_cleaning.py</code>  | NAV: pd.to_datetime, nav>0 filter, dedup on (amfi_code,date), resample('D').ffill() for holidays. Transactions: type mapping, KYC validation, annual_income_lakh>0. Performance: expense_ratio range check + anomaly flag. |
| <code>fetch_and_save(name, code)</code>   | <code>live_nav_fetch.py</code> | GET mfapi.in/{code} with 10s timeout, parses meta+data JSON, saves to data/raw/{name}_{code}_nav.csv.                                                                                                                      |

### Running the Pipeline

```
# From repo root:
python scripts/etl_pipeline.py

# Or run individual steps:
python data_ingestion.py # Anomaly scan
python data_cleaning.py # Per-table clean
python live_nav_fetch.py # Fetch live NAV
```

#### Key Design Decisions

- No hard-coded paths — pathlib.Path used throughout; root detected by checking cwd().name
- NAV weekend handling — resample('D').ffill() applied after sort+dedup to fill Saturday/Sunday/holiday gaps
- Logging — Python logging module with INFO level; each file load and write is logged
- Error isolation — per-file try/except so one bad CSV never aborts the whole run

## 5. D2 — SQLite Database Schema

[sql/schema.sql](#) | [sql/load\\_db.py](#) | [sql/queries.sql](#)

The database follows a **star schema**: two dimension tables (`dim_fund`, `dim_date`) and four fact tables. All fact tables carry a foreign key to `dim_fund.amfi_code`. The `.db` file is excluded from version control via `.gitignore`; the schema is reproduced here for reproducibility.

### dim\_fund — Fund Master Dimension

| Column                          | Type    | Constraint  | Description                       |
|---------------------------------|---------|-------------|-----------------------------------|
| <code>amfi_code</code>          | INTEGER | PRIMARY KEY | Unique AMFI scheme identifier     |
| <code>scheme_name</code>        | TEXT    | NOT NULL    | Full scheme name                  |
| <code>fund_house</code>         | TEXT    |             | AMC name                          |
| <code>category</code>           | TEXT    |             | Equity / Debt / Hybrid            |
| <code>sub_category</code>       | TEXT    |             | Large Cap, ELSS, etc.             |
| <code>plan</code>               | TEXT    |             | Regular / Direct                  |
| <code>launch_date</code>        | TEXT    |             | ISO date string                   |
| <code>benchmark</code>          | TEXT    |             | e.g. NIFTY100                     |
| <code>expense_ratio_pct</code>  | REAL    |             | Annual TER (%)                    |
| <code>exit_load_pct</code>      | REAL    |             | Exit load (%)                     |
| <code>min_sip_amount</code>     | REAL    |             | Minimum SIP (INR)                 |
| <code>fund_manager</code>       | TEXT    |             | Manager name                      |
| <code>risk_category</code>      | TEXT    |             | Low / Moderate / High / Very High |
| <code>sebi_category_code</code> | TEXT    |             | SEBI classification code          |

### dim\_date — Date Dimension

| Column                  | Type    | Description                     |
|-------------------------|---------|---------------------------------|
| <code>date_id</code>    | TEXT    | PRIMARY KEY — YYYY-MM-DD format |
| <code>day</code>        | INTEGER | Day of month                    |
| <code>month</code>      | INTEGER | Month (1–12)                    |
| <code>year</code>       | INTEGER | Calendar year                   |
| <code>quarter</code>    | INTEGER | Quarter (1–4)                   |
| <code>is_weekend</code> | INTEGER | 1 if Saturday or Sunday, else 0 |

### Fact Tables

| Table                          | Key Columns                                                                                                                                                                                                                                            | Notable Constraints                                   |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <code>fact_nav</code>          | <code>amfi_code</code> , <code>date_id</code> , <code>nav</code>                                                                                                                                                                                       | <code>CHECK(nav &gt; 0)</code>                        |
| <code>fact_transactions</code> | <code>investor_id</code> , <code>date_id</code> , <code>amfi_code</code> , <code>transaction_type</code> , <code>amount_inr</code> , <code>state</code> , <code>kyc_status</code>                                                                      | FK to <code>dim_fund</code> and <code>dim_date</code> |
| <code>fact_performance</code>  | <code>amfi_code</code> , <code>return_1yr_pct</code> , <code>return_3yr_pct</code> , <code>return_5yr_pct</code> , <code>sharpe_ratio</code> , <code>alpha</code> , <code>beta</code> , <code>max_drawdown_pct</code> , <code>expense_ratio_pct</code> | FK to <code>dim_fund</code>                           |

| Table    | Key Columns                                                 | Notable Constraints               |
|----------|-------------------------------------------------------------|-----------------------------------|
| fact_aum | date_id, fund_house, aum_lakh_crore, aum_crore, num_schemes | Units in column names as required |

Analytical Queries (sql/queries.sql)

| #   | Query Description                             | Tables Used       |
|-----|-----------------------------------------------|-------------------|
| Q1  | Top 5 fund houses by total AUM                | fact_aum          |
| Q2  | Average monthly NAV per fund                  | fact_nav          |
| Q3  | SIP year-on-year growth (window function LAG) | fact_transactions |
| Q4  | Transaction volume and amount by state        | fact_transactions |
| Q5  | Funds with expense ratio < 1%                 | fact_performance  |
| Q6  | Top 5 funds by 1-year return                  | fact_performance  |
| Q7  | Monthly SIP inflow trend                      | fact_transactions |
| Q8  | Redemption vs SIP ratio                       | fact_transactions |
| Q9  | Highest 5-year return with low expense ratio  | fact_performance  |
| Q10 | KYC status breakdown                          | fact_transactions |

6. **D3 — Exploratory Data Analysis**

*notebooks/EDA\_Analysis.ipynb | 11 charts in reports/charts/*

The EDA notebook covers all 10 processed datasets and produces 11 saved charts (a mix of interactive Plotly HTML and static Matplotlib/Seaborn PNG). Each chart is accompanied by a documented finding in markdown cells.

**Charts & Findings**

| Chart                       | File                        | Finding                                                                                                                               |
|-----------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| NAV Trends (line)           | 01_nav_trends.html          | Large-cap schemes showed 12–15% CAGR (2022–2026); sharp 2023 bull run post-COVID recovery. Interactive — hover for fund-level detail. |
| AUM Growth (bar)            | 02_aum_growth.png           | SBI Mutual Fund manages Rs.12.5L Cr+ AUM — nearly 2x the second-largest AMC (HDFC). Top 5 AMCs control ~65% of industry AUM.          |
| SIP Trend (line)            | 03_sip_trend.html           | Monthly SIP inflows crossed Rs.20,000 Cr in 2024; active accounts grew from 4.9 Cr to 7.4 Cr in three years.                          |
| Category Heatmap            | 04_category_heatmap.png     | Small Cap had the highest inflows in Apr–Jun 2024; Debt funds saw outflows post budget announcement.                                  |
| Demographics (pie+box)      | 05_demographics.png         | 25–35 age group is the largest SIP investor cohort; 30–40-year bracket contributes highest lumpsum amounts.                           |
| Geographic (bar+choropleth) | 06_geographic.png           | Maharashtra, Karnataka, Delhi top states by SIP amount; Tier-2 cities growing fastest YoY.                                            |
| Folio Growth (line)         | 07_folio_growth.html        | Industry folio count approaching 20 Cr; equity folios at 70% share. Consistent uptrend with no reversal.                              |
| Correlation Matrix          | 08_correlation_matrix.png   | Large-cap funds highly correlated ( $r > 0.85$ ); small-cap–debt correlation near zero confirms diversification benefit.              |
| Sector Donut (pie)          | 09_sector_donut.html        | Financials (24%), IT (18%) and Consumer Goods (12%) dominate equity portfolio holdings.                                               |
| Return Distribution         | 10_daily_return_dist.png    | Daily returns roughly normal with negative skew; small-cap tails heavier (kurtosis $> 4$ ) confirming higher tail risk.               |
| Benchmark Comparison        | 11_benchmark_comparison.png | Top-5 funds (by scorecard) outperformed NIFTY50 and NIFTY100 over the 3-year analysis window.                                         |

## 7. D4 — Performance Metrics

*notebooks/Performance\_Analysis.ipynb | 3 output CSVs*

All metrics are computed from daily NAV returns for 40 fund schemes. The risk-free rate proxy is 6.5% per annum (RBI repo rate), expressed as a daily rate (RF\_DAILY = 0.065 / 252). Benchmark for Alpha/Beta is NIFTY100.

### Metric Formulas

| Metric         | Formula                                                                                        | Annualisation                         | Output CSV         |
|----------------|------------------------------------------------------------------------------------------------|---------------------------------------|--------------------|
| Daily Return   | $r_t = (\text{NAV}_t / \text{NAV}_{\{t-1\}}) - 1$                                              | —                                     | —                  |
| CAGR           | $(\text{NAV}_{\text{end}} / \text{NAV}_{\text{start}})^{(252 / n_{\text{trading\_days}})} - 1$ | 252 trading days — NOT calendar years | fund_scorecard.csv |
| Sharpe Ratio   | $(\text{mean}(r - \text{RF\_daily}) / \text{std}(r)) \times \text{sqrt}(252)$                  | $\text{sqrt}(252)$ annualisation      | fund_scorecard.csv |
| Sortino Ratio  | $(\text{mean}(r - \text{RF\_daily}) / \text{std}(r[r < 0])) \times \text{sqrt}(252)$           | Downside deviation only               | fund_scorecard.csv |
| Alpha          | Intercept of OLS: $r_{\text{fund}} = \alpha + \beta \times r_{\text{NIFTY100}}$                | Annualised x 252                      | alpha_beta.csv     |
| Beta           | Slope of OLS regression above                                                                  | —                                     | alpha_beta.csv     |
| Max Drawdown   | $\min(\text{NAV}_t / \text{cummax}(\text{NAV})) - 1$                                           | —                                     | fund_scorecard.csv |
| Tracking Error | $\text{std}(r_{\text{fund}} - r_{\text{benchmark}}) \times \text{sqrt}(252)$                   | Annualised                            | tracking_error.csv |

### Top Performers (fund\_scorecard.csv)

The composite score weights: Sharpe rank 40%, CAGR rank 25%, Alpha rank 20%, Drawdown rank 10%, Expense ratio rank 5%.

| Rank | Scheme Name                          | CAGR 3Y (%) | Sharpe | Max DD (%) | Score |
|------|--------------------------------------|-------------|--------|------------|-------|
| 1    | Mirae Asset Large Cap — Regular      | 34.00       | 1.45   | -11.3      | 88.9  |
| 2    | ICICI Pru Bluechip — Direct          | 32.49       | 1.03   | -12.6      | 82.0  |
| 3    | ICICI Pru Midcap — Regular           | 31.78       | 1.18   | -18.2      | 80.5  |
| 4    | Kotak Flexicap — Regular             | 29.58       | 1.31   | -13.0      | 79.2  |
| 5    | HDFC Mid-Cap Opportunities — Regular | 32.44       | 1.09   | -16.2      | 78.1  |

### Most/Least Risky Funds (VaR 99%)

| Scheme                     | Category | VaR 95% (%) | VaR 99% (%) | CVaR 99% (%) |
|----------------------------|----------|-------------|-------------|--------------|
| ABSL Small Cap — Regular   | Equity   | -2.60       | -3.71       | -4.21        |
| DSP Small Cap — Regular    | Equity   | -2.35       | -3.68       | -4.08        |
| SBI Small Cap — Direct     | Equity   | -2.69       | -3.61       | -4.00        |
| — — —                      | — — —    | — — —       | — — —       | — — —        |
| ABSL Liquid — Regular      | Debt     | -0.03       | -0.05       | -0.06        |
| ICICI Pru Liquid — Regular | Debt     | -0.02       | -0.05       | -0.06        |
| Kotak Liquid — Regular     | Debt     | -0.03       | -0.05       | -0.06        |

## 8. D6 — Advanced Analytics

*notebooks/05\_advanced\_analytics.ipynb | var\_analysis.csv | cohort\_retention.png*

The advanced analytics notebook is structured in three self-contained sections: Value at Risk, Investor Cohort Analysis, and a Fund Recommender System.

### 8.1 Value at Risk (VaR)

Historical simulation VaR is computed at both 95% and 99% confidence levels for all 40 funds using the empirical return distribution. Conditional VaR (Expected Shortfall) is also reported as the mean loss beyond the VaR threshold.

```
returns = daily_nav.pct_change().dropna()
var_95 = np.percentile(returns, 5) # 5th percentile
var_99 = np.percentile(returns, 1) # 1st percentile
cvar_95 = returns[returns <= var_95].mean() # Expected Shortfall
```

#### VaR Interpretation

- VaR 99% = -3.5% means: on any given trading day, there is a 1% chance of losing more than 3.5%
- CVaR (Expected Shortfall) = average loss on the days that do exceed the VaR threshold — captures true tail risk
- Small-cap funds show VaR 99% of -3.6% to -3.7%; liquid debt funds are as low as -0.05%
- Output saved to data/processed/var\_analysis.csv with columns: amfi\_code, var\_95\_pct, var\_99\_pct, cvar\_95\_pct, cvar\_99\_pct

### 8.2 Investor Cohort Analysis

Investors are grouped by their first transaction month (cohort). For each cohort, month-by-month retention is computed as the percentage of the original cohort that remained active in each subsequent month. Results are visualised as a heatmap (cohort\_retention.png).

```
txn['cohort_month'] = txn.groupby('investor_id')['transaction_date']
    .transform('min').dt.to_period('M')
txn['months_since_join'] = (txn['txn_month'] - txn['cohort_month']).apply(lambda x: x.n)
cohort_pct = cohort_table.divide(cohort_table.iloc[:, 0], axis=0) * 100
```

#### Cohort Findings

- 5,000 investors tracked from Jan 2024 to May 2025 across 17 monthly cohorts
- Jan 2024 cohort (earliest) shows 34–36% retention through months 1–11 — relatively stable
- Newer cohorts (2025) show lower retention, partly due to truncated observation window
- Month-1 retention averages ~25% across all cohorts — indicating high first-month churn
- Chart saved to reports/charts/cohort\_retention.png

### 8.3 Fund Recommender System

A rule-based scoring recommender ingests the fund\_scorecard.csv and ranks funds by a weighted composite of normalised Sharpe ratio, CAGR, max-drawdown resilience, and expense ratio. Three risk profiles are supported.

| Risk Profile       | Weight: Sharpe | Weight: CAGR | Weight: Drawdown | Weight: Expense |
|--------------------|----------------|--------------|------------------|-----------------|
| Low (conservative) | 40%            | 35%          | 20%              | 5%              |
| Medium (balanced)  | 40%            | 35%          | 15%              | 10%             |
| High (aggressive)  | 35%            | 45%          | 10%              | 10%             |

Top Recommendations by Risk Profile

| Risk | Rank | Scheme                               | CAGR 3Y | Sharpe | Score |
|------|------|--------------------------------------|---------|--------|-------|
| Low  | 1    | Mirae Asset Large Cap — Regular      | 34.0%   | 1.45   | 0.931 |
| Low  | 2    | Kotak Flexicap — Regular             | 29.6%   | 1.31   | 0.872 |
| Low  | 3    | SBI Bluechip — Regular               | 30.5%   | 1.21   | 0.842 |
| High | 1    | Mirae Asset Large Cap — Regular      | 34.0%   | 1.45   | 0.964 |
| High | 2    | Axis Midcap — Regular                | 35.1%   | 1.00   | 0.901 |
| High | 3    | HDFC Mid-Cap Opportunities — Regular | 32.4%   | 1.09   | 0.888 |

## 9. Data Dictionary

Full column-level reference for all processed tables

| Table / Column           | Type  | Description                                                                             |
|--------------------------|-------|-----------------------------------------------------------------------------------------|
| 01_fund_master           | —     | —                                                                                       |
| amfi_code                | int   | Unique AMFI scheme code — primary key across all tables                                 |
| scheme_name              | str   | Full scheme name as registered with AMFI                                                |
| fund_house               | str   | Asset Management Company (AMC) name                                                     |
| category                 | str   | Broad asset class: Equity, Debt, Hybrid                                                 |
| sub_category             | str   | SEBI sub-category e.g. Large Cap, ELSS, Gilt                                            |
| risk_grade               | str   | Low / Low to Moderate / Moderate / High / Very High                                     |
| expense_ratio_pct        | float | Total Expense Ratio (%) — validated range 0.1–2.5                                       |
| 02_nav_history           | —     | —                                                                                       |
| date                     | date  | NAV date; forward-filled for weekends and holidays                                      |
| nav                      | float | Net Asset Value in INR — must be > 0                                                    |
| 03_aum_by_fund_house     | —     | —                                                                                       |
| aum_crore                | float | AUM in INR Crore (10 million)                                                           |
| aum_lakh_crore           | float | AUM in INR Lakh Crore (1 trillion) — for industry totals                                |
| 07_scheme_performance    | —     | —                                                                                       |
| return_1yr_pct           | float | 1-year trailing return (%)                                                              |
| return_3yr_pct           | float | 3-year trailing return (%)                                                              |
| return_5yr_pct           | float | 5-year trailing return (%)                                                              |
| expense_ratio_flag       | str   | 'anomaly' if outside 0.1–2.5% range, else 'ok'                                          |
| 08_investor_transactions | —     | —                                                                                       |
| transaction_type         | str   | SIP / Lumpsum / Redemption (standardised)                                               |
| amount_inr               | float | Transaction amount in INR — must be > 0                                                 |
| state                    | str   | Investor's Indian state                                                                 |
| city_tier                | str   | Tier 1 / Tier 2 / Tier 3 city classification                                            |
| kyc_status               | str   | KYC Verified / Pending / Rejected                                                       |
| annual_income_lakh       | float | Annual income in INR Lakh — rows with <= 0 removed                                      |
| fund_scorecard.csv       | —     | —                                                                                       |
| cagr_3yr                 | float | 3-year CAGR using $(252/n\_trading\_days)$ exponent                                     |
| sharpe_ratio             | float | Annualised Sharpe: $\text{mean}(\text{excess\_ret})/\text{std} \times \text{sqrt}(252)$ |
| sortino_ratio            | float | Annualised Sortino: uses downside std only                                              |
| max_drawdown             | float | Most negative trough-from-peak value (%)                                                |
| score                    | float | Composite weighted rank score 0–100                                                     |
| alpha_beta.csv           | —     | —                                                                                       |
| alpha                    | float | OLS intercept vs NIFTY100, annualised x252                                              |
| beta                     | float | OLS slope vs NIFTY100                                                                   |
| r_squared                | float | Coefficient of determination of regression                                              |

| Table / Column   | Type  | Description                                                        |
|------------------|-------|--------------------------------------------------------------------|
| var_analysis.csv | —     | —                                                                  |
| var_95_pct       | float | Historical VaR at 95% confidence (5th percentile of daily returns) |
| var_99_pct       | float | Historical VaR at 99% confidence (1st percentile)                  |
| cvar_95_pct      | float | Conditional VaR / Expected Shortfall beyond 95% VaR                |
| cvar_99_pct      | float | Conditional VaR / Expected Shortfall beyond 99% VaR                |

## 10. Known Issues & Pending Work

### Resolved Issues

| Issue                                | Resolution                                                                                                                              |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| .db files committed to git           | Added *.db to .gitignore; run 'git rm --cached bluestock_mf.db sql/bluestock_mf.db' to untrack                                          |
| CAGR using calendar years            | Fixed — now uses $(252/n\_trading\_days)$ exponent in both Performance and Advanced notebooks                                           |
| queries.sql column name mismatches   | Fixed — aum_crore, amount_inr, return_1yr_pct, return_5yr_pct now match schema exactly                                                  |
| Hard-coded Windows path in notebooks | EDA notebook uses pathlib detection; Performance notebook still has E:\PROJECTS\ in cell 0 (cosmetic — path is re-detected from cell 2) |
| Duplicate data_injection.py (typo)   | Removed from repo                                                                                                                       |

### Pending Work

| Deliverable               | What is needed                                                                                                                               | Priority        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| D5 — Dashboard (.pbix)    | Build 4-page Power BI dashboard: Fund Overview, NAV Trends, Investor/SIP Analysis, Performance Scorecard. Each page needs minimum 2 slicers. | HIGH — 20 marks |
| D7 — Final Report (PDF)   | Professional PDF report covering: Executive Summary, Methodology, EDA Insights, Performance Findings, Advanced Analytics, Recommendations.   | HIGH — 15 marks |
| D7 — Presentation (.pptx) | 10–15 slide deck mirroring the report structure with charts embedded.                                                                        | HIGH — 15 marks |
| EDA notebook cleanup      | Delete redundant path-setup cells (cells 1, 3, 4) and re-run clean so all outputs are fresh.                                                 | MEDIUM          |
| Performance notebook path | Cell 0 still has hard-coded BASE = r'E:\PROJECTS\...' — replace with pathlib detection.                                                      | LOW             |
| B1 — Cron ETL             | Add crontab entry or APScheduler wrapper around live_nav_fetch.py to run every weekday at 8 PM.                                              | BONUS           |
| B2 — Streamlit app        | Build streamlit app as Power BI alternative.                                                                                                 | BONUS           |
| B3 — Monte Carlo          | 5-year NAV projection with uncertainty bands in 05_advanced_analytics.ipynb or separate notebook.                                            | BONUS           |
| B4 — Markowitz frontier   | scipy.optimize.minimize for Efficient Frontier for 5 selected funds.                                                                         | BONUS           |
| B5 — HTML email report    | smtplib + jinja2 weekly performance email generator.                                                                                         | BONUS           |

Document generated: June 2025 | Repository: [github.com/Vindhya1110/mutual-fund-dashboard](https://github.com/Vindhya1110/mutual-fund-dashboard) | Author: Vindhya